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Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$$V_{CC} = 8\text{ V}; R_{E1} = 0.68\text{ k}\Omega; R_{E2} = 4.7\text{ k}\Omega; R_P = 6.8\text{ k}\Omega; R_S = 10\text{ k}\Omega; C_S = 1\text{ }\mu\text{F}.$$

$Q_{2,3,4}$ (PNP BJT): $V_{BEp} = -0.7\text{ V}$; $\beta_{Fp} = \beta_{Op} = 150$; $V_{CEsatp} = -0.2\text{ V}$; $r_{Op} = 280\text{ k}\Omega$;

Q_1 (NPN BJT): $V_{BE n} = 0.7\text{ V}$; $\beta_{Fn} = \beta_{On} = 200$; $V_{CEsatn} = 0.2\text{ V}$; $r_{On} = +\infty$;

$$V_T = 26\text{ mV}.$$

Considering all capacitors to be equivalent to open circuits, the current mirror transistors Q_3 and Q_4 to be identical and operating at the same temperature and neglecting Early effect ($\rightarrow I_{C4} = I_{C3}$), determine:

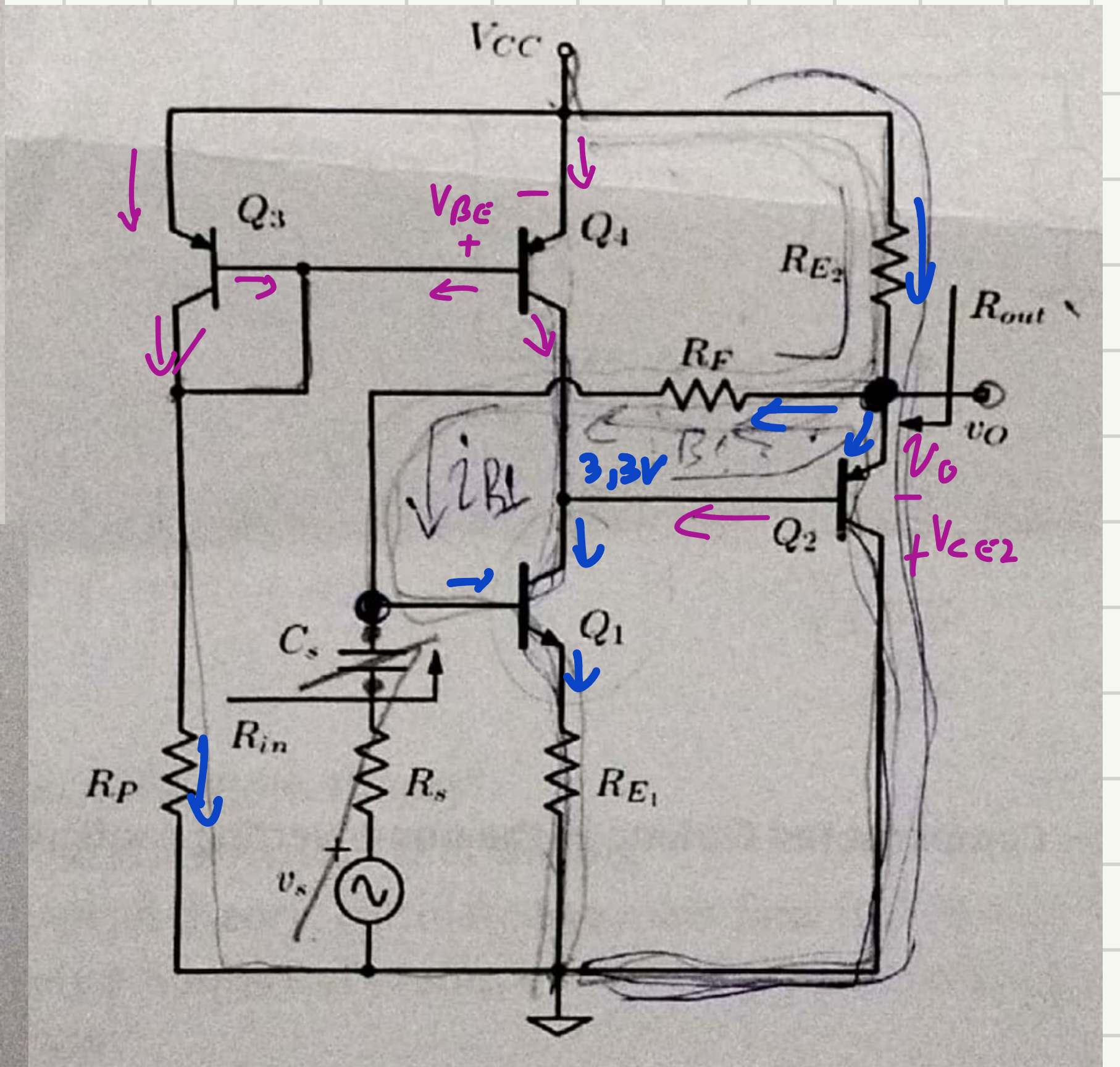
1. the value of R_F that ensures $V_O = -V_{CE2}$ is biased at 4 V ;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), capacitor C_S is equivalent to a short circuit and taking into account the current mirror loading effect on Q_1 , determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance R_{in} indicated in the figure;
5. the output resistance R_{out} indicated in the figure.

Finally, considering the effect of capacitor C_S :

6. the low frequency bandwidth limit f_L of the amplifier.



15:00

$$I_{C4} + I_{B2} - I_{C1} = 0$$

$$I_{B2} = I_{C1} - I_{C4}$$

$$I_{B34} = \frac{I_{RP}}{\beta_{F3} + 2} = 7.1063\text{ }\mu\text{A}$$

$$I_{C3} = I_{C4} = \beta_F I_{B34} = 1.06\text{ mA}$$

$$I_{RE2} = \frac{V_{CC} - V_O}{R_{E2}} = 0.851\text{ mA} = I_{C2} + I_{B1}$$

$$I_{C2} = (1 + \beta_{F2}) I_{B2} = (1 + \beta_{F2}) (I_{C1} - I_{C4}) = (1 + \beta_{F2}) (\beta_{F1} I_{B1} - I_{C4})$$

so

$$\begin{aligned} I_{RE2} &= (1 + \beta_{F2}) (\beta_{F1} I_{B1} - I_{C4}) + I_{B1} \\ &= (1 + \beta_{F2}) \beta_{F1} I_{B1} - (1 + \beta_{F2}) I_{C4} + I_{B1} \\ &= I_{B1} ((1 + \beta_{F2}) \beta_{F1} + 1) - (1 + \beta_{F2}) I_{C4} \end{aligned}$$

$$I_{B1} = \frac{I_{RE2} + (1 + \beta_{F2}) I_{C4}}{(1 + \beta_{F2}) \beta_{F1} + 1} = 5.316\text{ }\mu\text{A} \quad I_{C1} = \beta_{F1} I_{B1} = 1.065\text{ mA}$$

$$R_F = \frac{V_O - V_{BE1} - R_{E1} (1 + \beta_F) I_{B1}}{I_{B1}} = 483\text{ k}\Omega$$

$$V_{CE1} = V_0 + V_{BE1} - R_{E1} I_{E1} = 2,57 \text{ V}$$

$$I_{E1} = 1,0685 \text{ mA}$$

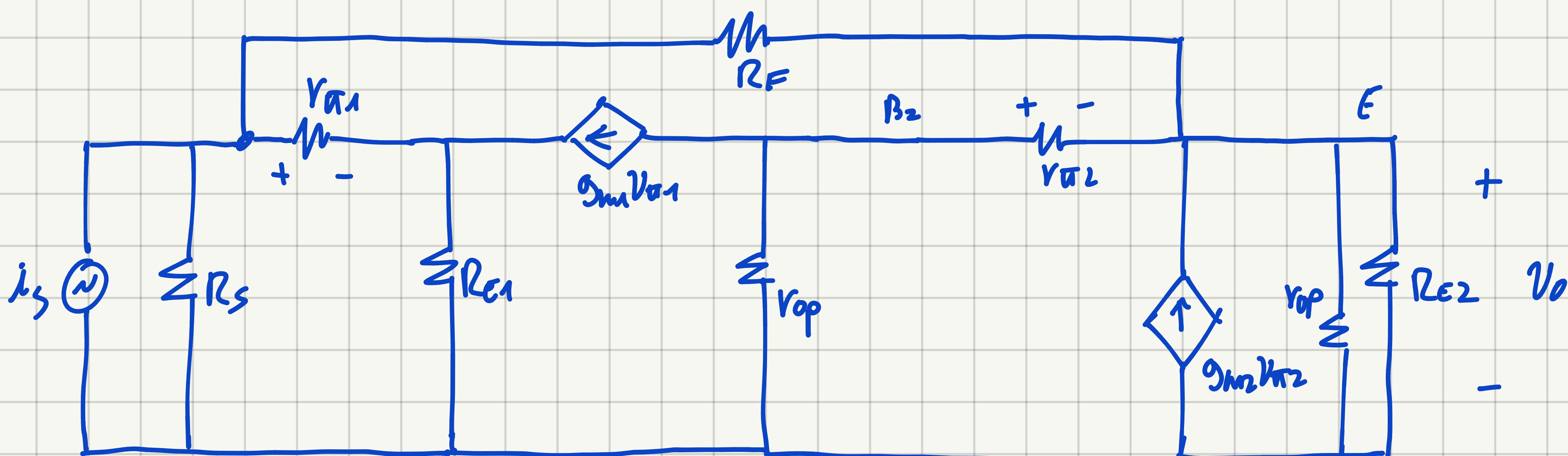
$$V_{CE2} = -4 \text{ V}$$

$$I_{C2} = I_{E2} - I_{B1} = 0,845 \text{ mA}$$

$$Q_1(2,57 \text{ V}, 1,06 \text{ mA}) \quad Q_2(-4 \text{ V}, 0,845 \text{ mA})$$

$$2) A_V = V_0/V_s \quad (\text{account for mirror loading})$$

20 min op point



$$g_{m1} = \frac{I_{C1}}{V_T} = 40,77 \text{ mS}^{-1}$$

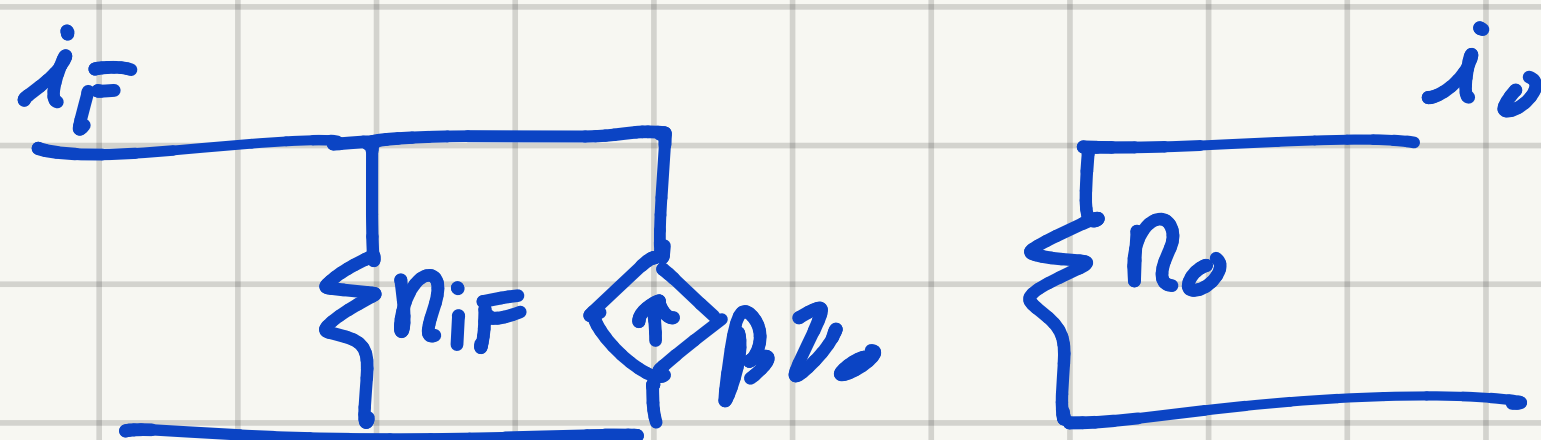
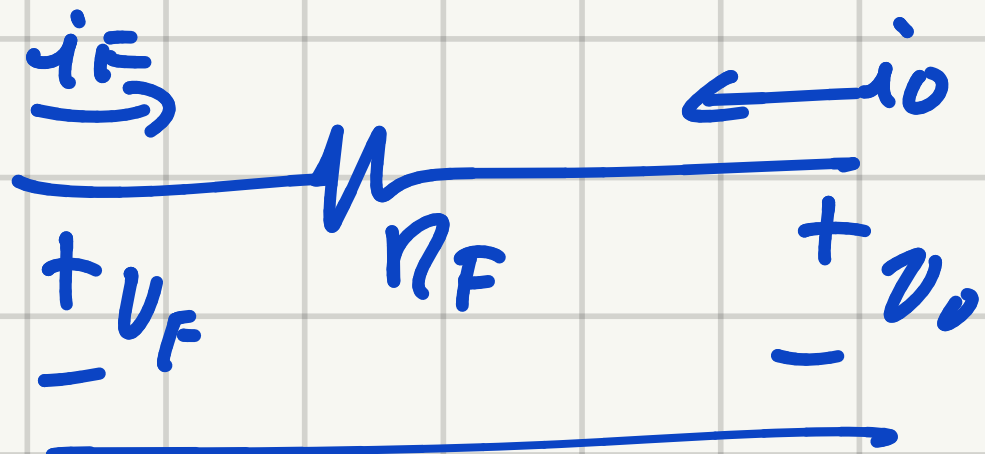
$$g_{m2} = 32,5 \text{ mS}^{-1}$$

$$r_{\pi 1} = 4,9 \text{ k}\Omega$$

$$r_{\pi 2} = 4,61 \text{ k}\Omega$$

We have voltage sampling current mixing

$$\beta \text{ is } i_F = f(V_0) = \begin{cases} i_F = \beta V_0 + V_F/R_{iF} \\ i_0 = V_0/R_0 + \end{cases}$$



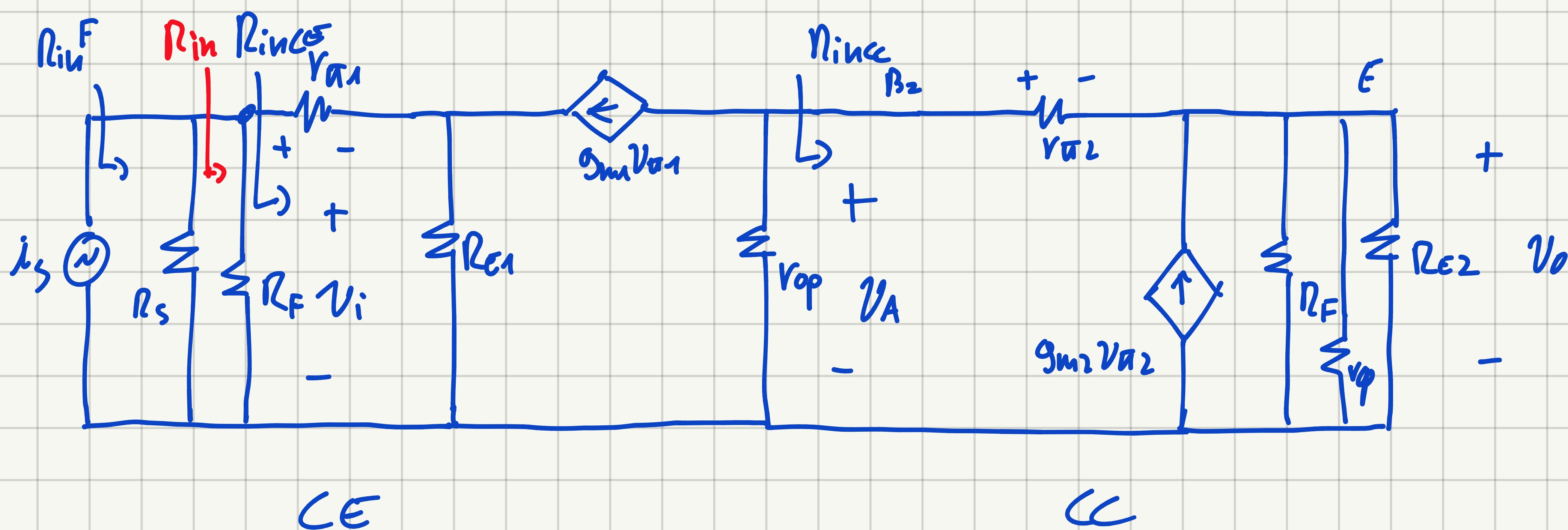
$$\beta = \frac{V_0}{i_F} \Big|_{V_{iF}=0} = -\frac{1}{R_F}$$

$$R_{iF} = \frac{V_F}{i_F} \Big|_{V_0=0} = R_F$$

$$R_{oF} = \frac{V_0}{i_0} \Big|_{V_F=0} = R_F$$

easy: Kill output feedback (short V_0) $\rightarrow$ you see R_F from input

kill input feedback (short V_s) $\rightarrow$ you see R_F from the output



$$A_v^o = \frac{v_o}{i_s} = A_{vcc} \cdot A_{vce} = \frac{v_o}{v_A} \cdot \frac{v_A}{v_i} \cdot \frac{v_i}{i_s}$$

$$R_x = r_{op} \parallel R_F \parallel R_{E2} = 4,58 \text{ k}\Omega$$

$$R_{incc} = r_{\pi 2} + (1 + \beta_{o2}) R_{E2} = 141,6 \text{ k}\Omega$$

$$R_{incc} = r_{\pi 2} + (1 + \beta_{o2}) R_x = 696,2 \text{ k}\Omega$$

$$R_s \parallel R_F = 9,8 \text{ k}\Omega$$

$$A_{vce2} = \frac{v_A}{v_s} = \frac{\beta_{o1} (r_{op} \parallel R_{incc})}{r_{\pi 1} + (1 + \beta_{o1}) R_{E1}} = -282$$

$$\frac{v_i}{i_s} = R_s \parallel R_F \parallel R_{incc} = 9,16 \text{ k}\Omega$$

$$A_v^{OL} = -2565,8 \text{ k}\Omega$$

$$1 + \beta A_v^{OL} = 6,3$$

$$A_v^F = \frac{A_v^{OL}}{1 + \beta A_v^{OL}} = 407,22 \text{ k}\Omega$$

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 8 \text{ V}$; $R_{E1} = 0.68 \text{ k}\Omega$; $R_{E2} = 4.7 \text{ k}\Omega$; $R_P = 6.8 \text{ k}\Omega$; $R_s = 10 \text{ k}\Omega$; $C_s = 1 \mu\text{F}$.

$Q_{1,3,4}$ (PNP BJT): $V_{BEp} = -0.7 \text{ V}$; $\beta_{Fp} = \beta_{Op} = 150$; $V_{CEsatp} = -0.2 \text{ V}$; $r_{op} = 280 \text{ k}\Omega$;

Q_2 (NPN BJT): $V_{BE n} = 0.7 \text{ V}$; $\beta_{Fn} = \beta_{On} = 200$; $V_{CEsatn} = 0.2 \text{ V}$; $r_{on} = +\infty$;

$V_T = 26 \text{ mV}$.

$$R_{in}^{OL} = R_s \parallel R_F \parallel R_{incc} = 9,16 \text{ k}\Omega$$

$$R_{in}^F = R_{in}^{OL} / (1 + \beta) = 1,45 \text{ k}\Omega$$

$$R_{out}^{OL} = R_{E2} \parallel r_{op} \parallel R_F \parallel (r_{\pi 2} + r_{op}) = 4,5 \text{ k}\Omega \quad R_{out}^F = R_{out}^{OL} / (1 + \beta) = 714 \Omega$$

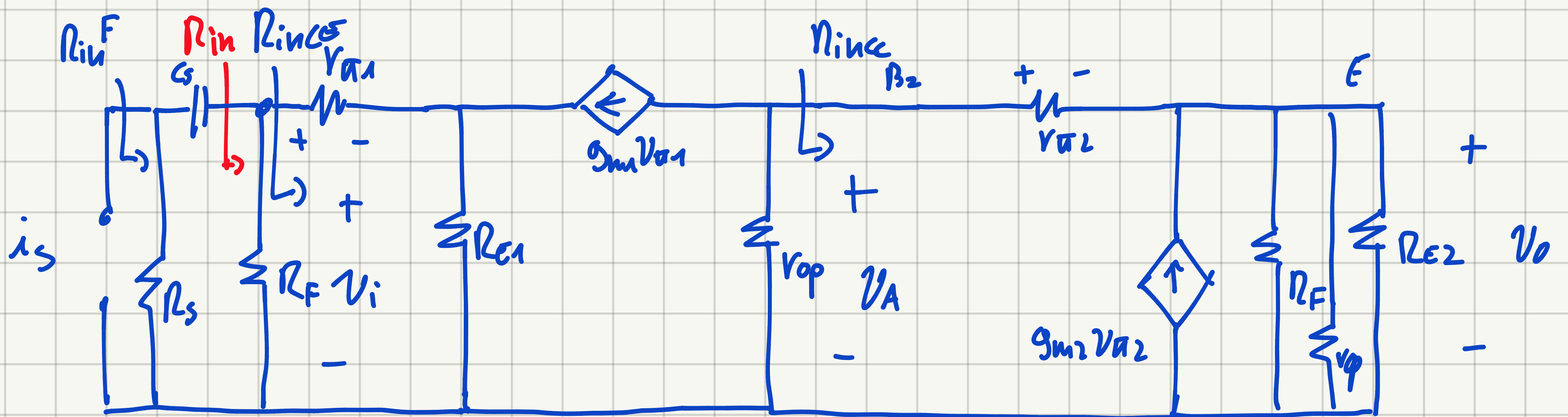
4) R_{in} and R_{out} of the figure

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_s}} = 1,7 \text{ k}\Omega$$

$$R_{out} = R_{out}^F = 714 \Omega$$

5) f_L due to $C_S = 1\mu F$ is affected by feedback

$$f_L = \frac{1}{2\pi} \cdot \frac{1}{C_S R_S^{sc}}$$



$$R_S^{sc} = R_s + R_{in} = 11,7 k\Omega$$

$$f_L^{OL} = 13,6 Hz$$

$$f_L^F = \frac{f_L^{OL}}{1+T} = 2,15 Hz$$

current mixing increase
 R_{in} so $1/R$ decrease.

Es 2

Determine:

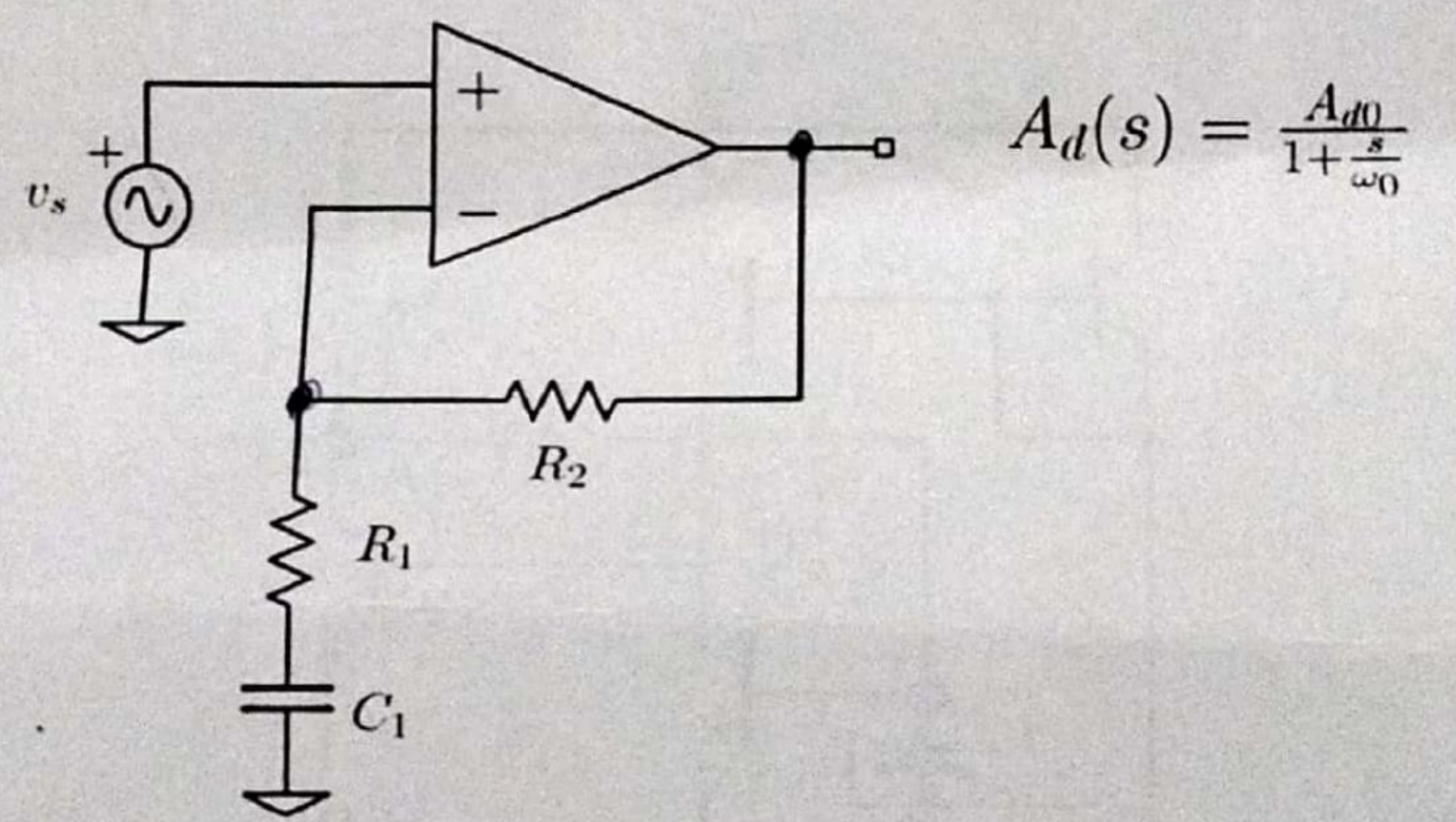
1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the amplifier bandwidth and *phase margin*.

Then, determine also:

3. the value of capacitor C_1 that makes the phase margin of the circuit equal to 60° ;
4. the amplifier bandwidth with the modified C_1 value.

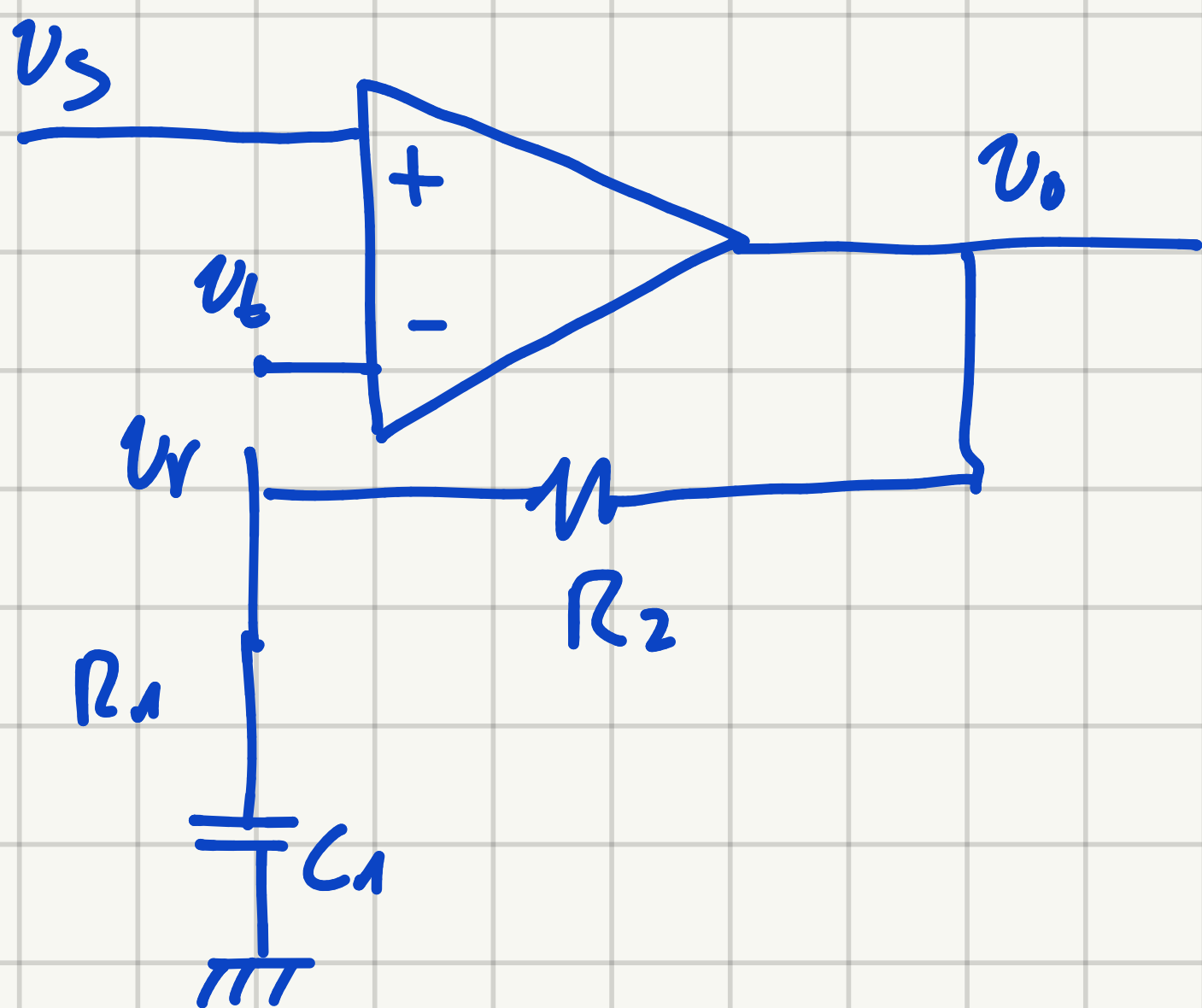
$$R_1 = 1 \text{ k}\Omega; R_2 = 68 \text{ k}\Omega; C_1 = 1 \text{ nF}.$$

$$\text{OpAmp: } A_{d0} = 10^5 \text{ [V/V]}, \omega_0 = 10 \text{ rad/s}.$$



$$1) A_v = \frac{v_o}{v_s} = 1 + \frac{R_2}{R_1} = 69$$

2) PM and BW



$$A_d = \frac{A_{d0}}{1 + s/\omega_0}$$

$$A_{d0} = 20 \log 10^5 = 100 \text{ dB}$$

$$\beta + 1 : \frac{1 + sC_1R_1}{sC_1}$$

get β from D.L.1

$$v_r = A_d v_t \cdot \frac{1 + sC_1R_1}{sC_1} = A_d v_t \frac{1 + sC_1R_1}{1 + sC_1(R_2 + R_1)}$$

$$\beta(s) = \frac{1 + sC_1R_1}{1 + sC_1(R_2 + R_1)}$$

$$\omega_z = \frac{1}{C_1R_1} = 10^6 \text{ rad/s}$$

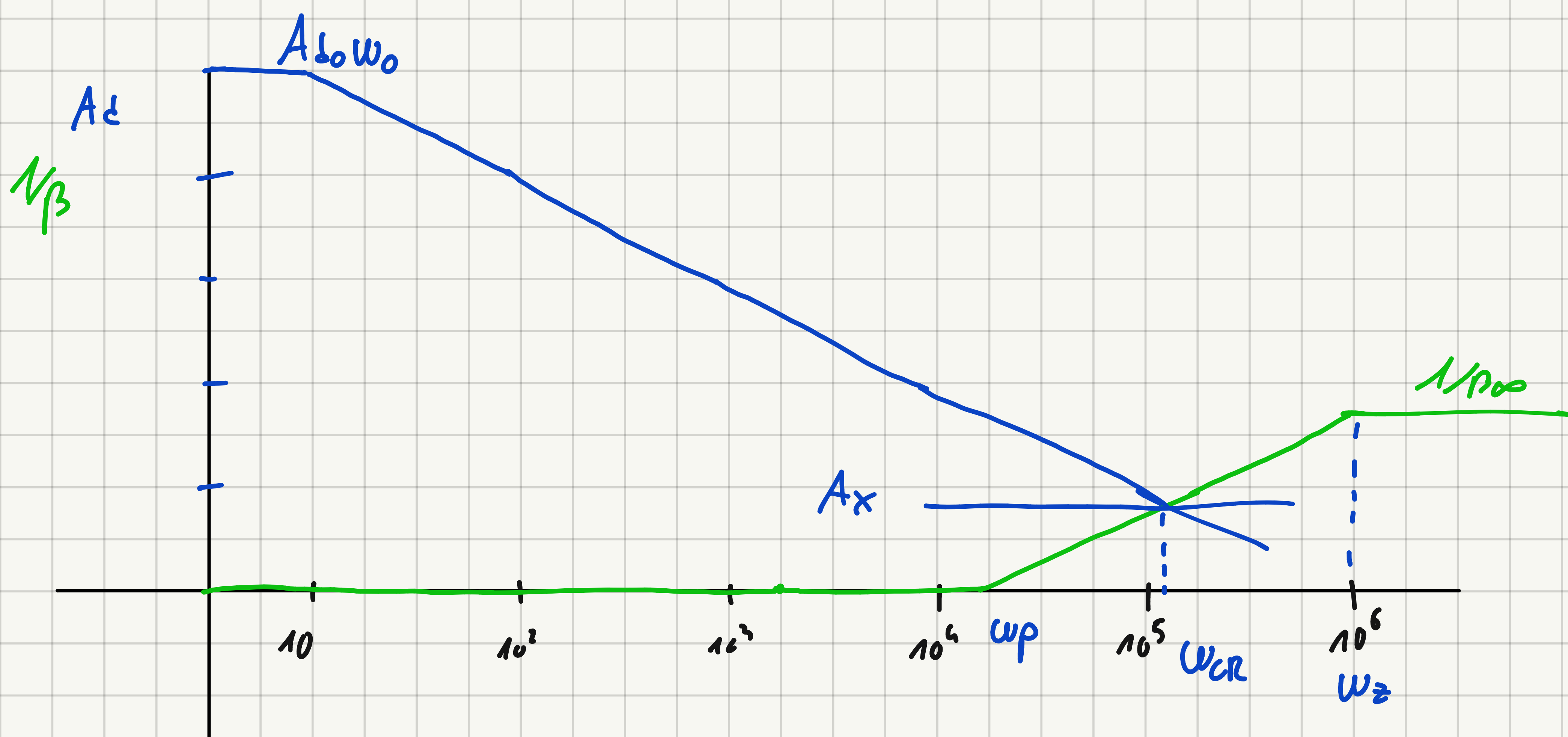
$$\frac{1}{\beta(s)} = \frac{1 + sC_1(R_2 + R_1)}{1 + sC_1R_1}$$

$$\omega_p = \frac{1}{C_1(R_2 + R_1)} = 1,45 \cdot 10^4 \text{ rad/s}$$

$$\frac{1}{\beta(0)} = 1 = 0 \text{ dB}$$

$$\frac{1}{\beta(\infty)} = \frac{C_1(R_2 + R_1)}{C_1R_1} = \frac{R_2 + R_1}{R_1} = 69$$

$$\frac{1}{\beta(\infty)} \text{ dB} = 36,7 \text{ dB}$$



$$\begin{cases} A_0 \omega_0 = A_x \omega_{cn} \\ \frac{1}{\omega_p} = \frac{A_x}{\omega_{cn}} \end{cases} \quad A_x = \frac{\omega_{cn}}{\omega_p}$$

$$A_0 \omega_0 = \frac{\omega_{cn}^2}{\omega_p} \quad \omega_{cn} = \sqrt{A_0 \omega_0 \omega_p} = 1,2 \cdot 10^5$$

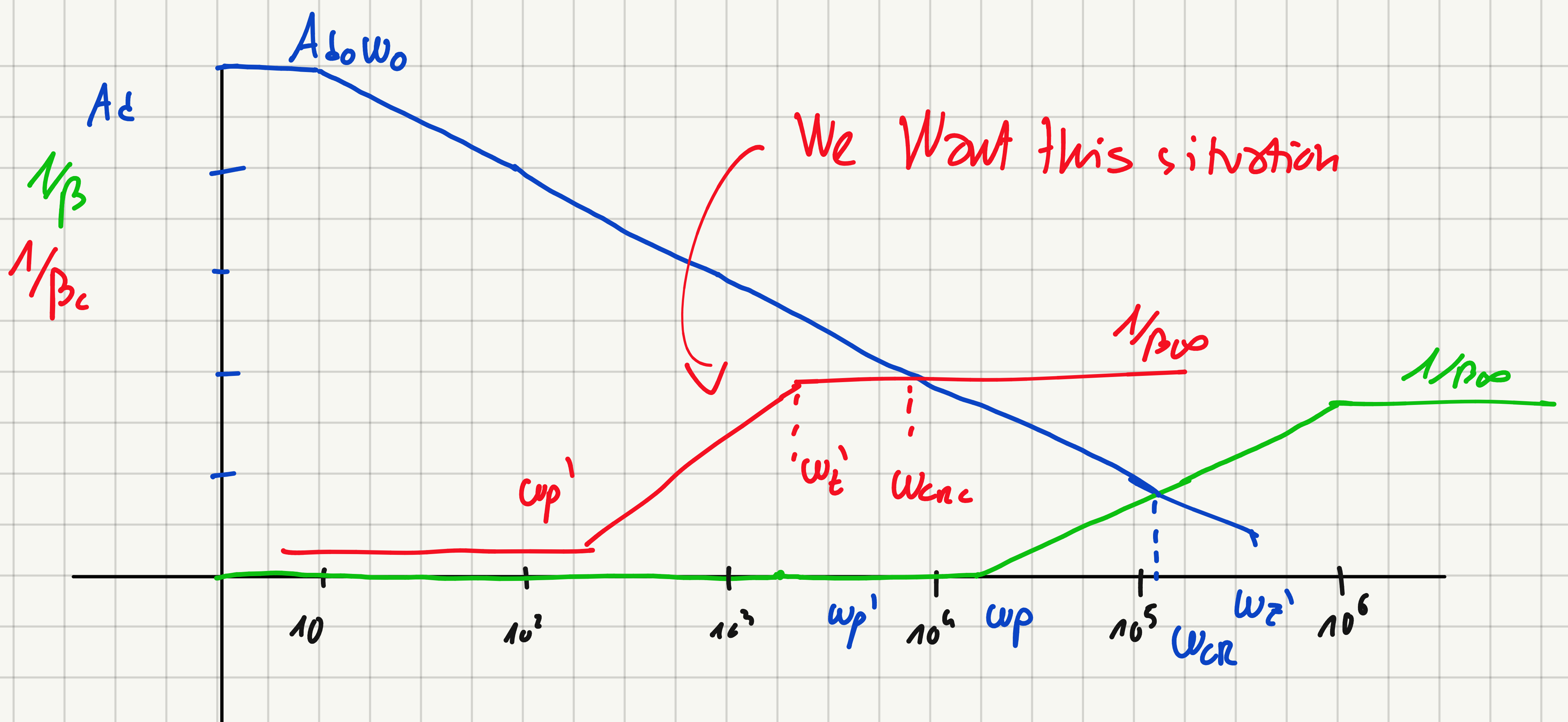
$$\varphi_m = 180 - \arctan\left(\frac{\omega_{cn}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cn}}{\omega_p}\right) + \arctan\left(\frac{\omega_{cn}}{\omega_z}\right) = 13,73^\circ$$

3) Value of C_1 that makes $\varphi_m = 60^\circ$

you have to move it bigger so $\omega_z = \omega_{cn}/h$

$$\omega_z = \frac{1}{C_1 R_1} = \frac{\omega_{cn}}{h}$$

you have to evaluate again the bandwidth to dimension C_1



$$A_{d0}\omega_0 = \frac{1}{\beta_{00}} \cdot \omega_{cr}$$

$$\omega_{cr} = A_{d0}\omega_0 \beta_{00} = 15 \cdot 10^3 \text{ rad/s}$$

$$\omega_{zc} = \frac{1}{C_1 R_1} = \frac{\omega_{cr}}{4}$$

$$C_1 = \frac{4}{R_1 \omega_{cr}} = 266 \text{ nF}$$

$$\omega_{pc} = \frac{1}{C_1 (R_1 + R_2)} = 54,34 \text{ rad/s}$$

$$\omega_{zc} = \frac{\omega_{cr}}{4} = 3,75 \cdot 10^3 \text{ rad/s}$$

$$\varphi_m' = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_{p'}}\right) + \arctan\left(\frac{\omega_{cr}}{\omega_{z'}}\right) = 76,2^\circ$$

process for compensation.

$$\varphi_m = 60^\circ \rightarrow \omega_z = \frac{\omega_{cr}}{4} \text{ AFTER compensation}$$

→ get ω_{cr} from the bode → get ω_{zc} (after imposing $\omega_z = \omega_{cr}/4$)

